Use of digital magnetometer for determination of temperature of deposition (i.e. to distinguish between block and ash flow vs. cold lahar):

(1) use a field compass to draw a horizontal line (water line) on the rock of interest in place in the outcrop. Also draw a north arrow on the rock.

(2) Repeat for at least 12 samples, roughly equant samples work best.
(3) Remove samples from outcrop
(4) Set up field magnetometer. Set the small magnetic sensor on a stable nonmagnetic surface, such that you can move your samples close enough to touch the sensor. The sensor must be at least one foot from the instrument to prevent interference. Here, the sensor is mounted on a wooden stake.

(5) Zero the readout (this removes the need to align your sensor with the current magnetic field).
(6) To test the setup, you can use a brunton compass. Freeze the magnetic arrow on your compass in place when pointed toward north. Approach the sensor with the compass north arrow toward the arrow on the sensor. The digital readout should read a positive value. Turn the compass around and do the same and it should be equally negative. If you tilt the compass, the response should decrease.
(7) Select one sample, orient the horizontal line with horizontal and point the north arrow toward the arrow on the sensor.

(8) Record the response. Tilt the rock to see if the response increases or decreases. Turn the sample around and do the same.

Rocks emplaced in the past 780,000 years should have normal polarity with near horizontal inclination. This means that blocks should have their north direction roughly parallel to north on the sensor (positive gamma reading during the first measurement) and should not increase in gamma response as you tilt the sample up or down at the same distance from the magnetic sensor (roughly zero inclination). When you turn the sample around, the gamma reading should be negative and again not increase as you tilt the sample.

This instrument is well suited to differentiate between hot block and ash flows and cool lahars. All juvenile blocks in a block and ash flow should record the same direction of magnetization. Blocks in a lahar have rotated (been remobilized) since they originally cooled and locked in a magnetization direction, so their orientations should be in many directions.